TSM 099 Final Exam - Practice 1

Version 202610122020.31 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Find the slope-intercept equation through these two points.

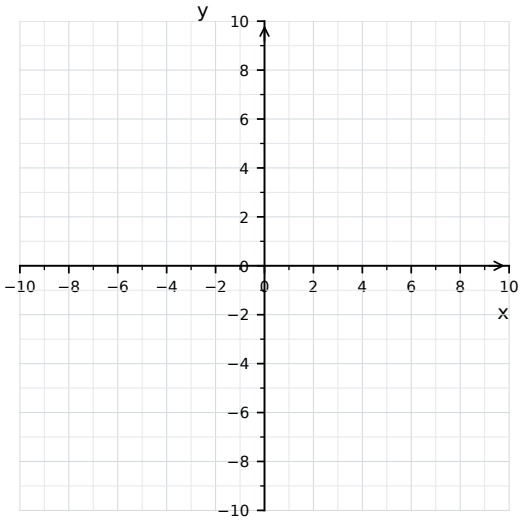
$(2,\ -1);\ (4,\ 0)$

2. Factor completely.

$6x^2 - 5x - 6$

3. Find the intercepts and graph the line on the grid.

$2x + y = 6$



4. Solve by taking square roots.

$3(x-1)^2 = 27$

5. Add and simplify.

$\frac{2}{x^2 - 2x} + \frac{3}{x^2 - 4}$

6. Solve using the quadratic formula. Give exact simplified solutions.

$2x^2 = 4 - x$

7. Solve by factoring.

$x^2 + 2x = 8$

8. Solve and check in the original equation.

$\frac{8}{2x-6} + \frac{2}{x-3} = 2$

9. Multiply and simplify.

$$\frac{x^2-4}{x^2-2x-35} \times \frac{x+5}{x^2+2x}$$

10. Evaluate at x = -2; show substitution.

$$(x+3)(2x-2)$$

Student answer sheet

TSM 099 Final Exam | Practice 1

Version 202610122020.31 | Name: _____

Question 1

Equation

10 pts

Question 2

Factors

10 pts

Question 3

Intercepts

4 pts

Graph (booklet)

6 pts

Question 4

Solutions

10 pts

Question 5

Expression

10 pts

Student answer sheet

TSM 099 Final Exam | Practice 1

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Question 6

Solutions

10 pts

Question 7

Solutions

10 pts

Question 8

Solution/check

10 pts

Question 9

Expression

10 pts

Question 10

Value

10 pts

Answer key

TSM 099 Final Exam | Practice 1

Version 202610122020.31 | Name: _____

Question 1

Equation	$y = \frac{x}{2} - 2$

10 pts

Question 2

Factors	$(2x - 3)(3x + 2)$

10 pts

Question 3

Intercepts	$(3,\ 0);\ (0,\ 6)$

4 pts

Graph (booklet)	$y = 6 - 2x$

6 pts

Question 4

Solutions	$x = -2,\ 4$

10 pts

Question 5

Expression	$\frac{5x + 4}{x(x - 2)(x + 2)}$

10 pts

Answer key

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Question 6

Solutions	$x = -\frac{1}{4} + \frac{\sqrt{33}}{4},\ \ -\frac{\sqrt{33}}{4} - \frac{1}{4}$

10 pts

Question 7

Solutions	$x = -4,\ 2$

10 pts

Question 8

Solution/check	$x = 6$

10 pts

Question 9

Expression	$\frac{x-2}{x(x-7)}$

10 pts

Question 10

Value	-6

10 pts